Aufgabe 1: Netzmasken

Bestimmen Sie für die folgenden IPv4-Adressen und Präfixlängen jeweils die **Subnetzadresse (Netz-ID)** und die **Hostadresse (Host-ID)** im angegebenen Subnetz.

Interaktiver Simulator (IPv4-Subnetting): Simulator ↗

Bitweise Zerlegung, Maskierung und Adressberechnung interaktiv im Browser nachvollziehen.

#	Gegebene IP / CIDR	Subnetzadresse	Hostadresse	Simulator
1	`10.79.108.192/28`	**`10.79.108.192`**	**`0.0.0.0`**	**Simulator ↗**
2	`160.34.216.41/24`	**`160.34.216.0`**	**`0.0.0.41`**	**Simulator ↗**
3	`194.140.235.98/22`	**`194.140.232.0`**	**`0.0.3.98`**	**Simulator ↗**
4	`70.144.241.168/26`	**`70.144.241.128`**	**`0.0.0.40`**	**Simulator ↗**
5	`36.62.54.161/24`	**`36.62.54.0`**	**`0.0.0.161`**	**Simulator ↗**
6	`118.34.184.194/21`	**`118.34.184.0`**	**`0.0.0.194`**	**Simulator ↗**
7	`90.152.200.147/23`	**`90.152.200.0`**	**`0.0.0.147`**	**Simulator ↗**
8	`126.236.168.167/28`	**`126.236.168.160`**	**`0.0.0.7`**	**Simulator ↗**
9	`120.25.104.175/29`	**`120.25.104.168`**	**`0.0.0.7`**	**Simulator ↗**
10	`243.115.3.123/27`	**`243.115.3.96`**	**`0.0.0.27`**	**Simulator ↗**

Interaktiver Simulator (IPv4-Subnetting): Simulator ↗

Bitweise Zerlegung, Maskierung und Adressberechnung interaktiv im Browser nachvollziehen.

#	Gegebene IP / CIDR	Subnetzadresse	Hostadresse	Simulator
11	`66.64.53.18/21`	**`66.64.48.0`**	**`0.0.5.18`**	**Simulator ↗**
12	`236.10.205.157/28`	**`236.10.205.144`**	**`0.0.0.13`**	**Simulator ↗**
13	`17.199.131.101/26`	**`17.199.131.64`**	**`0.0.0.37`**	**Simulator ↗**
14	`170.102.133.128/24`	**`170.102.133.0`**	**`0.0.0.128`**	**Simulator ↗**
15	`185.128.24.102/23`	**`185.128.24.0`**	**`0.0.0.102`**	**Simulator ↗**
16	`141.48.173.70/27`	**`141.48.173.64`**	**`0.0.0.6`**	**Simulator ↗**
17	`165.239.154.199/21`	**`165.239.152.0`**	**`0.0.2.199`**	**Simulator ↗**
18	`144.146.157.100/21`	**`144.146.152.0`**	**`0.0.5.100`**	**Simulator ↗**
19	`154.158.62.98/22`	**`154.158.60.0`**	**`0.0.2.98`**	**Simulator ↗**
20	`255.203.71.146/25`	**`255.203.71.128`**	**`0.0.0.18`**	**Simulator ↗**

Interaktiver Simulator (IPv4-Subnetting): Simulator ↗

Bitweise Zerlegung, Maskierung und Adressberechnung interaktiv im Browser nachvollziehen.

#	Gegebene IP / CIDR	Subnetzadresse	Hostadresse	Simulator
21	`142.31.142.7/22`	**`142.31.140.0`**	**`0.0.2.7`**	**Simulator ↗**
22	`79.66.160.180/29`	**`79.66.160.176`**	**`0.0.0.4`**	**Simulator ↗**
23	`201.199.146.123/22`	**`201.199.144.0`**	**`0.0.2.123`**	**Simulator ↗**
24	`18.123.38.113/28`	**`18.123.38.112`**	**`0.0.0.1`**	**Simulator ↗**
25	`23.226.192.253/26`	**`23.226.192.192`**	**`0.0.0.61`**	**Simulator ↗**
26	`189.92.223.28/25`	**`189.92.223.0`**	**`0.0.0.28`**	**Simulator ↗**
27	`2.216.124.116/28`	**`2.216.124.112`**	**`0.0.0.4`**	**Simulator ↗**
28	`85.235.192.189/26`	**`85.235.192.128`**	**`0.0.0.61`**	**Simulator ↗**
29	`121.22.231.143/26`	**`121.22.231.128`**	**`0.0.0.15`**	**Simulator ↗**
30	`190.49.174.54/27`	**`190.49.174.32`**	**`0.0.0.22`**	**Simulator ↗**

Interaktiver Simulator (IPv4-Subnetting): Simulator ↗

Bitweise Zerlegung, Maskierung und Adressberechnung interaktiv im Browser nachvollziehen.

#	Gegebene IP / CIDR	Subnetzadresse	Hostadresse	Simulator
31	`57.100.65.168/27`	**`57.100.65.160`**	**`0.0.0.8`**	**Simulator ↗**
32	`123.1.75.109/28`	**`123.1.75.96`**	**`0.0.0.13`**	**Simulator ↗**
33	`102.245.239.221/29`	**`102.245.239.216`**	**`0.0.0.5`**	**Simulator ↗**
34	`20.102.14.134/23`	**`20.102.14.0`**	**`0.0.0.134`**	**Simulator ↗**
35	`252.137.193.159/22`	**`252.137.192.0`**	**`0.0.1.159`**	**Simulator ↗**
36	`251.79.244.166/29`	**`251.79.244.160`**	**`0.0.0.6`**	**Simulator ↗**
37	`198.188.4.20/24`	**`198.188.4.0`**	**`0.0.0.20`**	**Simulator ↗**
38	`2.46.28.227/26`	**`2.46.28.192`**	**`0.0.0.35`**	**Simulator ↗**
39	`136.175.106.220/24`	**`136.175.106.0`**	**`0.0.0.220`**	**Simulator ↗**
40	`249.39.221.196/27`	**`249.39.221.192`**	**`0.0.0.4`**	**Simulator ↗**

Interaktiver Simulator (IPv4-Subnetting): Simulator ↗

Bitweise Zerlegung, Maskierung und Adressberechnung interaktiv im Browser nachvollziehen.

#	Gegebene IP / CIDR	Subnetzadresse	Hostadresse	Simulator
41	`110.46.57.88/27`	**`110.46.57.64`**	**`0.0.0.24`**	**Simulator ↗**
42	`61.66.248.84/23`	**`61.66.248.0`**	**`0.0.0.84`**	**Simulator ↗**
43	`29.110.115.49/22`	**`29.110.112.0`**	**`0.0.3.49`**	**Simulator ↗**
44	`187.117.201.2/26`	**`187.117.201.0`**	**`0.0.0.2`**	**Simulator ↗**
45	`115.48.156.102/27`	**`115.48.156.96`**	**`0.0.0.6`**	**Simulator ↗**
46	`132.37.160.159/22`	**`132.37.160.0`**	**`0.0.0.159`**	**Simulator ↗**
47	`92.211.191.64/24`	**`92.211.191.0`**	**`0.0.0.64`**	**Simulator ↗**
48	`154.211.167.32/29`	**`154.211.167.32`**	**`0.0.0.0`**	**Simulator ↗**
49	`102.84.30.97/24`	**`102.84.30.0`**	**`0.0.0.97`**	**Simulator ↗**
50	`170.223.206.121/26`	**`170.223.206.64`**	**`0.0.0.57`**	**Simulator ↗**